

MINI PROJECT DOCUMENTATION

STUDENT EMERGENCY NETWORK

A Smart Campus Emergency Assistance and Alert System

Project Type: B.Tech CSE Mini Project

Development Platform: Base44 AI

Live Application: vivid-campus-safe-link.base44.app

Prepared for GitHub Project Documentation

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1. Project Overview

Student Emergency Network is a web-based campus safety application designed to help students quickly report emergencies and connect with authorized campus responders. The application provides a simple SOS mechanism, emergency reporting, location sharing, emergency contacts, notifications, and responder/admin dashboards.

The project was developed using **Base44 AI**, an AI-powered application development platform. The platform was used to create the web application interface, application logic, data management, authentication, and workflow components with natural-language instructions.

2. Problem Statement

During an emergency on a college campus, students may not know whom to contact or may lose valuable time searching for help. Traditional communication methods such as phone calls, messaging groups, or manually contacting campus staff can be slow and difficult to coordinate.

The proposed system addresses this problem by providing a centralized digital platform through which students can quickly raise an emergency alert and provide relevant information such as emergency type, description, and location.

3. Objectives

The main objectives of the Student Emergency Network are:

- Provide a quick and simple SOS emergency mechanism.
- Allow students to report different types of campus emergencies.
- Share location information only when explicitly permitted by the student.
- Notify authorized responders about active emergencies.
- Allow responders to accept and update emergency status.
- Provide administrators with emergency monitoring and analytics.
- Maintain emergency records for future reference.
- Improve coordination between students, responders, and campus administration.

4. Proposed Solution

The Student Emergency Network consists of three major user interfaces: Student, Responder, and Administrator. A student can create an SOS alert or submit a detailed emergency report. The alert is stored in the application database and becomes visible to authorized responders. A responder can accept the alert, update the response status, and mark it as resolved. Administrators can monitor emergency activity and manage users, responders, contacts, and reports.

5. Key Features

- **One-Tap SOS:** Quickly initiate an emergency alert after confirmation.
- **Emergency Reporting:** Report medical, accident, fire, security, harassment, missing-person, or other incidents.
- **Location Sharing:** Capture location when the student grants browser permission.

- **Responder Dashboard:** View, accept, and manage active emergencies.
- **Admin Dashboard:** Monitor emergency statistics and records.
- **Emergency Contacts:** Quickly access configured campus emergency contacts.
- **Notifications:** Display relevant alert and status updates.
- **Safety Center:** Provide campus safety information and guidance.
- **Emergency History:** Students can review their own previous reports.
- **Responsive UI:** Designed for desktop and mobile screens.

6. Technology and Tools Used

Component	Technology / Tool	Purpose
Application Development	Base44 AI	AI-assisted web application development
Frontend	Base44-generated web UI	Responsive user interface
Backend / Data	Base44 application backend	Data storage and application workflows
Authentication	Base44 authentication	Role-based user access
Location	Browser Geolocation API / permission-based location	Emergency location sharing
Hosting	Base44 deployment	Live web application

7. System Modules

7.1 Student Module

The student module provides the main emergency functions. Students can log in, access the dashboard, send an SOS alert, submit emergency reports, share location when needed, access emergency contacts, view notifications, and check previous reports.

7.2 Responder Module

Responders receive access to active emergency information relevant to their role. They can accept an alert, view the available location information, update response status, and resolve the emergency.

7.3 Administrator Module

The administrator module provides centralized monitoring and management. Administrators can manage students and responders, configure emergency contacts, view reports, and monitor emergency statistics.

7.4 Safety Center

The Safety Center provides general information such as first-aid basics, fire safety, emergency preparedness, and instructions for reporting incidents.

8. System Workflow

Student → SOS/Report → Database → Responder Notification → Responder Accepts → Response Status Updates → Emergency Resolved

Typical workflow:

1. Student logs into the application.
2. Student selects SOS or Report Emergency.
3. Student confirms the SOS or enters emergency details.
4. If location is required, the browser requests permission.
5. The emergency record is stored in the application database.
6. Authorized responders can view the active alert.
7. A responder accepts the alert and updates its status.
8. The responder marks the incident as resolved after the response.

9. Database Design

The application uses structured entities to organize users and emergency information.

Entity	Important Fields
Users	User ID, name, email, role, phone, department
Emergency Alerts	Alert ID, student ID, type, severity, location, time, status, responder ID
Emergency Reports	Report ID, student ID, category, description, severity, location, status
Responders	Responder ID, name, contact, availability, status
Emergency Contacts	Contact ID, department, phone, availability
Notifications	Notification ID, user ID, message, type, time, read status

10. User Roles

Student: Create SOS alerts, report emergencies, share location when required, view emergency contacts, notifications, and personal report history.

Responder: Monitor active emergencies, accept alerts, view permitted emergency information, update response status, and resolve incidents.

Administrator: Manage users and responders, configure emergency contacts, monitor reports, and view analytics.

11. Base44 AI Development

Base44 AI was used as the primary development platform. The application requirements were described using natural-language prompts, after which the platform generated and configured application components. The development process focused on iterative prompting, reviewing generated screens and workflows, and refining the application according to the mini-project requirements.

The use of Base44 AI reduced the amount of manual coding required for the prototype and allowed the project to focus on application design, workflow logic, usability, and problem solving.

12. User Interface and Pages

The main pages of the application include:

- Landing Page
- Login / Registration
- Student Dashboard
- SOS Emergency Page
- Report Emergency Page
- Emergency Contacts
- My Reports
- Safety Center
- Responder Dashboard
- Admin Dashboard
- Notifications
- Profile / Settings

Screenshot placeholders for GitHub documentation:

1. Landing Page screenshot
2. Student Dashboard screenshot
3. SOS Emergency screenshot
4. Report Emergency screenshot
5. Responder Dashboard screenshot
6. Admin Dashboard screenshot

13. Advantages

- Faster emergency reporting within a campus environment.
- Simple and accessible user interface.
- Centralized emergency information.
- Role-based access for students, responders, and administrators.
- Permission-based location sharing.
- Emergency status tracking improves coordination.
- Easy to demonstrate and extend as a CSE mini project.

14. Limitations

- The system depends on internet/browser availability.
- Location sharing depends on device and browser permissions.
- The application is a campus-assistance prototype and does not replace official emergency services.
- Actual emergency contact numbers and responder infrastructure must be configured by the institution.
- Response time depends on the availability of authorized responders.

15. Future Enhancements

Future versions can include:

- Real-time map tracking for authorized responders.
- SMS and push notifications.
- Integration with official campus security systems.
- QR-based emergency identification.
- AI-assisted emergency classification and prioritization.
- Multilingual support.
- Offline emergency queueing with later synchronization.
- Advanced incident analytics and heat maps.
- Verified campus responder mobile application.

16. Testing

The application should be tested using functional, usability, security, and responsiveness test cases.

Test Case	Expected Result
Student Login	Valid user reaches student dashboard
Send SOS	Confirmation appears and alert is created
Emergency Report	Report is stored with required details
Location Permission	Location is captured only after permission
Responder Accept	Alert status changes to accepted/responding
Resolve Emergency	Alert is marked resolved and history updates
Role Access	Users see only permitted pages/data
Mobile UI	Pages remain usable on small screens

17. GitHub Repository Structure

Recommended repository structure:

```
Student-Emergency-Network/  
■■■■ README.md  
■■■■ documentation/  
■ ■■■■ Student_Emergency_Network_Documentation.pdf  
■■■■ screenshots/  
■ ■■■■ landing-page.png  
■ ■■■■ student-dashboard.png  
■ ■■■■ sos-page.png  
■ ■■■■ responder-dashboard.png  
■ ■■■■ admin-dashboard.png  
■■■■ LICENSE
```

If the Base44 project source is not exported, the GitHub repository can serve as the project documentation repository and include the live Base44 application link, screenshots, requirements, and this PDF.

18. How to Use the Application

1. Open the live application in a modern web browser.
2. Register or log in using an appropriate user role.
3. Students can use the SOS button for an emergency alert or submit a detailed report.
4. Grant browser location permission only when location sharing is needed.
5. Authorized responders can monitor and manage active alerts.
6. Administrators can monitor the overall emergency system.

19. Conclusion

The Student Emergency Network demonstrates how an AI-assisted development platform can be used to create a practical web application for campus safety. The project combines emergency reporting, SOS alerts, permission-based location sharing, responder coordination, notifications, and administrative monitoring in a single platform.

As a B.Tech CSE mini project, it provides a practical demonstration of web application development, database concepts, authentication, role-based access, responsive UI design, and AI-assisted software development using Base44 AI.

20. Project Link

Live Application: <https://vivid-campus-safe-link.base44.app>

Note: This project is intended as a campus emergency-assistance prototype. It should not be considered a replacement for official emergency services.